

A
PROJECT REPORT
ON
Jewellery Marketing and Saving Schemes App

Submitted in Partial Fulfillment of Requirement for the award of
degree of

BACHELOR OF TECHNOLOGY
COMPUTER SCIENCE AND ENGINEERING

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL
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Certificate

This is to certify that the project entitled, "**Jewellery Marketing and Saving Scheme App**" is

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It is a bonafide work carried out by these students under guidance of Prof. D. V. Karche. It has been accepted and approved for the partial fulfillment of the requirement of Dr. Babasaheb Ambedkar Technological University, Lonere, for the award of the degree of Bachelor of Technology (Computer Science and Engineering). This work and project report has not been earlier submitted to any other Institute or University for the award of any degree or diploma.

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Acknowledgments

The satisfaction and euphoria that accompany the successful completion of any task would be incomplete without the mention of people who made it possible. So, we acknowledged all those whose guidance and encouragement served as a beacon of light and crowned our efforts with success.

We have immense pleasure in expressing thanks to the Principal **Dr. A. D. Jadhav** , for providing all the facilities for the successful completion of the project.

Our profound sense of gratitude is due to our Head of the Department **Prof. P. M. Pondkule**, for constant encouragement, valuable guidance and also for the inspiration to pursue the project.

We also thanks to **Dr. V. K. Bhosle** Project Coordinator, for their guidance and for being a constant source of support and guided us in this Endeavour.

Last but not the least we are thankful to all faculty members and lab instructors without whose support at various stages, this project would not have materialized.

“Thanking You”

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Abstract

The project involves the development of a full-featured jewellery marketing and scheme-saving application designed to streamline product browsing, customer engagement, and gold saving scheme management. The platform provides a modern digital interface for customers to explore jewellery collections, compare prices, receive personalized recommendations, and access real-time product details. The system also digitizes traditional gold savings schemes, enabling users to enroll in monthly installment plans, track payments, view scheme benefits, and receive automated reminders through a secure and intuitive workflow. The application supports user authentication, digital record maintenance, automated notifications, and secure payment handling to ensure reliability and transparency. Jewellers benefit from a dedicated dashboard for product catalog management, scheme configuration, customer insights, promotional offers, and inventory updates. The platform is built with a modular architecture to support scalability, smooth UI rendering, and efficient backend operations, ensuring responsiveness across devices. By integrating marketing features, savings scheme automation, and customer relationship tools into a unified ecosystem, this project serves as a practical example of implementing a modern jewellery management platform. It emphasizes user experience, operational efficiency, digital transformation, and future-ready jewellery business workflows.

Keywords: Jewellery Marketing App, Gold Saving Schemes, Monthly Installment Plans, Digital Catalogue, Scheme Tracking, Customer Dashboard, Jeweller Dashboard, Product Management, Digital Payments, Real-Time Notifications, Jewellery Business Automation.

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CHAPTER 1: Introduction

In today's digital age, the integration of technology into traditional business models has become essential for enhancing efficiency, transparency, and customer satisfaction. The Jewellery industry, which has long been rooted in conventional practices, is now witnessing a significant shift towards digitalization. Customers today expect seamless online experiences, secure transactions, and easy access to investment options, especially when dealing with high-value assets such as gold, silver, and diamond Jewellery. However, existing jewelry marketing and saving schemes largely depend on manual operations, physical store interactions, and paper-based records, making them inefficient, time-consuming, and prone to errors. Jewelry saving schemes have been a popular method for customers to invest gradually and purchase jewelry at a later stage. These schemes typically involve monthly installments over a fixed period, after which customers can redeem their savings for jewelry purchases, often with added benefits such as discounts or bonus amounts. Despite their popularity, traditional saving schemes suffer from several drawbacks, including lack of transparency, difficulty in tracking payments, risk of mismanagement, and limited accessibility for customers who cannot frequently visit stores. Moreover, customers often face challenges in maintaining records of their payments and understanding scheme terms clearly, which can lead to confusion and mistrust. The project titled "Jewellery Marketing and Saving Schemes App" is designed to overcome these limitations by providing a comprehensive digital platform that combines jewelry marketing with systematic saving mechanisms. The primary objective of this application is to digitize and automate the entire process of enrolling in saving schemes, managing installments, and purchasing jewelry, thereby offering a more efficient and user-friendly solution. By leveraging modern web and mobile technologies, the system ensures that customers can access services anytime and from anywhere, eliminating the need for frequent physical visits to jewelry stores. One of the key features of the proposed system is its ability to provide real-time tracking of savings and transactions. Customers can easily view their installment history, remaining balance, maturity date, and benefits associated with their chosen scheme.

Automated notifications and reminders help users stay consistent with their payments, reducing the chances of missed installments. Additionally, the application offers a wide range of jewelry products with detailed descriptions, images, and pricing, enabling customers to make informed decisions. Security and reliability are critical aspects of the proposed system. The application incorporates secure authentication mechanisms and encrypted payment gateways to ensure safe trans-

actions. Digital records of all activities are maintained in a centralized database, reducing the risk of data loss and ensuring accuracy. This level of transparency not only enhances customer trust but also strengthens the credibility of jewelry businesses. Another important advantage of the system is its scalability and adaptability. The application can be extended to support multiple branches, large-scale jewelry chains, and even online marketplaces. It can also be integrated with financial services to provide additional investment options such as gold savings plans or digital gold purchases. By bridging the gap between traditional practices and modern technology, the system creates a flexible platform that can evolve with changing market demands. In addition to improving customer convenience, the application contributes to the overall digital transformation of the jewelry industry. It reduces dependency on manual processes, minimizes operational costs, and enhances efficiency for both customers and businesses. The system also promotes financial discipline among users by encouraging regular savings through structured schemes. The growing adoption of smartphones and internet services has significantly influenced customer expectations in terms of accessibility and convenience. Users now prefer platforms that allow them to perform transactions, track investments, and explore products from the comfort of their homes. In this context, a digital jewellery saving and marketing platform not only meets these expectations but also creates new opportunities for customer engagement and retention. By offering a user-friendly interface and seamless navigation, the application ensures that even non-technical users can easily utilize its features. In addition, the integration of modern technologies such as cloud computing and secure databases enhances the overall performance and reliability of the system. Cloud-based storage enables efficient handling of large volumes of data, while ensuring data availability and backup. This eliminates the risks associated with physical record maintenance and provides a scalable solution that can grow with increasing user demand. The system also supports real-time updates, ensuring that customers and administrators always have access to the latest information.

In conclusion, the “Jewellery Marketing and Saving Schemes App” represents a significant advancement in the digital transformation of the jewelry industry. It addresses key challenges associated with traditional systems, including inefficiency, lack of transparency, and limited accessibility. By providing a secure, user-friendly, and feature-rich platform, the application enhances the overall experience of jewelry purchasing and saving. It not only benefits customers by offering convenience and reliability but also empowers businesses with tools for growth and innovation. Ultimately, this project contributes to the development of a more efficient, transparent, and technologically driven jewelry ecosystem, paving the way for future advancements in the industry. From an organizational perspective, the application

enables Jewellery businesses to adopt a more strategic approach to marketing and customer management. Digital platforms allow businesses to analyze customer data, identify trends, and personalize their offerings accordingly. This data-driven approach enhances decision-making and helps businesses remain competitive in a rapidly evolving market. Additionally, online presence increases brand visibility and allows businesses to reach a wider audience beyond geographical limitations. Finally, the implementation of the Jewellery Marketing and Saving Schemes App contributes to the broader goal of digital transformation in traditional industries. It demonstrates how technology can be effectively utilized to improve existing systems, enhance user experience, and drive innovation. By bridging the gap between conventional practices and modern digital solutions, the project lays the foundation for future advancements and sets a benchmark for similar applications in the Jewellery sector.

CHAPTER 1. INTRODUCTION

Recent Case: Rise of Digital Gold Investment in the News:

The screenshot shows a news article from The Economic Times | wealth. The header includes the newspaper's name, 'English Edition', the date '30 April, 2026, 10:04 AM IST', and 'Today's ePaper'. Below the header is a promotional banner for ETPrime, offering a 15-day free trial. The article is marked as an 'ADVERTORIAL'. The title is 'Akshaya Tritiya Meets Smart Investing: The Rise of Digital Gold'. Below the title, it says 'ET Special' and 'Last Updated: Apr 18, 2026, 07:33:00 PM IST'. A 'Synopsis' section follows, describing Akshaya Tritiya as a day for prosperity that now embraces digital gold. Below the synopsis is an 'AI Briefing' section with a 'Listen' button. The main body of the article features a photograph of a family and text explaining that Akshaya Tritiya is an auspicious day for buying gold, and that digital gold offers a modern, safe, and flexible way to invest.

THE ECONOMIC TIMES | **wealth**
English Edition • | 30 April, 2026, 10:04 AM IST | Today's ePaper

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ADVERTORIAL

Akshaya Tritiya Meets Smart Investing: The Rise of Digital Gold

ET Special Last Updated: Apr 18, 2026, 07:33:00 PM IST

Synopsis
Akshaya Tritiya, a day for prosperity, now embraces digital gold. This modern investment allows easy online gold purchases, offering convenience and flexibility. Investors can start small, ensuring consistent savings. Digital gold blends tradition with smart financial planning for a secure future. It provides a safe and accessible way to honor this timeless gold tradition.

AI Briefing
Listen to this article in summarized format **Listen**

ET Spotlight
According to tradition, Akshaya Tritiya is viewed as an auspicious day for buying gold to attract prosperity and good fortune. Investing in digital gold has now become a modern method of purchasing gold without needing to go to the store, and it provides an opportunity for individuals to invest small amounts of money safely and consistently. Digital gold provides great flexibility for celebrating traditions while allowing individuals to take charge of their financial planning.

Akshaya Tritiya has long been regarded as an auspicious date for starting new ventures or making major purchases in India. One of the many customs associated with this event is the purchase of gold; it symbolises wealth, security and a stable investment.

Figure 1: Real-World Impact of Document Forgery

This news highlights that during Akshaya Tritiya, many individuals are shifting from traditional gold purchases to digital gold investment through mobile apps. Investors are now preferring small, regular investments with features like real-time tracking and secure transactions. This shift results in the growing demand for a system that ensures transparency, convenience, and efficient management of jewellery saving schemes.

CHAPTER 2: Literature Review

Table 1: Literature Review on Jewellery Marketing and Saving Schemes App

Sr. No.	Author & Year	Methods Used	Challenges	Applications
1	R. Sharma, P. Verma (2025)	Mobile-based digital gold investment platforms with real-time tracking and secure transactions.	Lack of trust in digital systems and user adoption challenges.	Digital gold saving apps, online jewellery investment platforms.
2	A. Gupta, S. Mehta (2024)	E-commerce integration for jewellery marketing using web and mobile applications.	Security issues in online transactions and data breaches.	Online jewellery stores, digital product catalogs and marketing platforms.
3	K. Patel, R. Singh (2024)	Secure payment gateway integration using UPI, encryption, and authentication techniques.	Data privacy concerns and risk of financial fraud.	Online payment systems for jewellery saving schemes and transactions.
4	M. Khan, S. Ali (2023)	Cloud-based data management system for storing customer records and scheme details.	Data loss risks and scalability issues in large systems.	Customer data storage, scheme tracking, and business management systems.
5	P. Desai, N. Joshi (2023)	Mobile application development with notification systems for saving schemes.	User engagement and retention difficulties.	Jewellery saving apps with automated reminders and alerts.

CHAPTER 2. LITERATURE REVIEW

Table 1: Literature Review on Jewellery Marketing and Saving Schemes App

Sr. No.	Author & Year	Methods Used	Challenges	Applications
6	S. Iyer, V. Nair (2023)	Digital marketing strategies using personalized recommendations and analytics.	Limited reach of traditional marketing methods.	Online jewellery promotion, targeted advertisements, customer engagement.
7	T. Roy, A. Das (2023)	Analysis of customer behavior in digital investment platforms and apps.	Resistance to shift from traditional to digital systems.	Digital investment platforms and user behavior analysis.
8	D. Kulkarni, R. Patil (2022)	Web-based inventory and sales management systems for jewellery businesses.	Manual errors and inefficiency in stock handling.	Jewellery inventory management and sales tracking systems.
9	N. Shah, P. Jain (2022)	Integration of SMS and mobile app notifications for payment reminders.	Missed payments and lack of timely communication.	Automated saving scheme reminders and customer notifications.
10	V. Reddy, K. Kumar (2022)	Hybrid system combining offline and online jewellery business models.	Complexity in system integration and maintenance.	Multi-channel jewellery business platforms and digital transformation.

CHAPTER 2. LITERATURE REVIEW

Table 1: Literature Review on Jewellery Marketing and Saving Schemes App

Sr. No.	Author & Year	Methods Used	Challenges	Applications
11	Chaffey, D. (2019)	Study of digital marketing strategies in e-commerce platforms focusing on customer engagement and product promotion. Businesses use SEO, social media, and targeted ads to attract customers.	High competition, maintaining customer trust, and low conversion rates.	Online jewellery marketing platforms, digital catalogs, and customer engagement systems.
12	Turban, E., King, D. (2020)	Comprehensive analysis of e-commerce systems including product management, inventory control, online transactions, and secure payment gateway integration. The study highlights the importance of user-friendly interfaces and efficient backend systems.	Security risks in online payments, data breaches, and lack of transparency in transaction processes.	Jewellery e-commerce websites, online selling platforms, and integrated business management systems

CHAPTER 2. LITERATURE REVIEW

Table 1: Literature Review on Jewellery Marketing and Saving Schemes App

Sr. No.	Author & Year	Methods Used	Challenges	Applications
13	Dahlberg, T., Mallat, N. (2008)	Research on mobile payment systems and digital wallets, focusing on fast, secure, and convenient financial transactions using smartphones. The study evaluates technologies like mobile banking and electronic payment methods.	Security concerns, lack of standardization, and low user trust in digital payment systems.	UPI payments, mobile wallets, and secure transaction systems used in jewellery saving scheme applications.
14	Kim, C., Mirusmonov, M. (2010)	Study on factors influencing user adoption of mobile applications including ease of use, trust, and convenience.	User resistance and concerns regarding privacy and security.	Mobile-based jewellery apps, saving scheme platforms, and customer-friendly applications.
15	Gefen, D., Karahanna, E. (2003)	Research on trust and user acceptance in online shopping environments. Focus on security, reliability, and user satisfaction.	Building trust in online systems and ensuring data protection.	Online jewellery purchasing systems, saving scheme applications, and customer trust models.

2.1 Technology Description

1.Flutter Framework:

Flutter is an open-source, cross-platform UI development framework developed by Google that enables the creation of high-performance mobile applications using a single codebase. It is widely used for building applications for Android, iOS, web, and desktop platforms. Flutter is based on a widget-driven architecture, where every element of the user interface is represented as a widget. These widgets can be combined and customized to create complex and visually appealing user interfaces. In the Jewellery Marketing and Saving Schemes App, Flutter is utilized to design a responsive and interactive user interface that allows users to browse jewellery products, enroll in saving schemes, and manage their transactions efficiently. One of the key features of Flutter is its hot reload functionality, which allows developers to instantly see changes in the application without restarting it, thereby speeding up the development process. Flutter also provides a rich set of pre-designed widgets and tools that help in creating modern and attractive user interfaces. The architecture of Flutter consists of three main layers: the framework layer, the engine layer, and the embedder layer. The framework layer includes UI components, animations, and gesture handling, while the engine layer is responsible for rendering, text layout, and graphics. T

Furthermore, Flutter supports advanced features such as state management, which helps in efficiently managing application data and updating the user interface dynamically. Popular state management techniques include Provider, Riverpod, and Bloc, which are used to handle complex application logic in a structured manner. Flutter also provides seamless integration with backend services such as Firebase, enabling real-time data synchronization, authentication, and cloud storage. In addition to performance and flexibility, Flutter provides excellent support for responsive design and adaptive layouts. This allows the application to function efficiently across different screen sizes and devices, ensuring a consistent user experience on smartphones and tablets. Flutter also supports internationalization and localization, enabling the application to cater to users from different regions and languages, which is particularly beneficial for expanding jewellery businesses to a wider audience. Overall, Flutter provides a powerful and efficient platform for developing scalable, maintainable, and feature-rich applications. Its flexibility, performance, rich UI capabilities, and strong ecosystem make it highly suitable for building the Jewellery Marketing and Saving Schemes App. The use of Flutter ensures a smooth, secure, and engaging user experience while supporting the long-term growth and adaptability of the application.

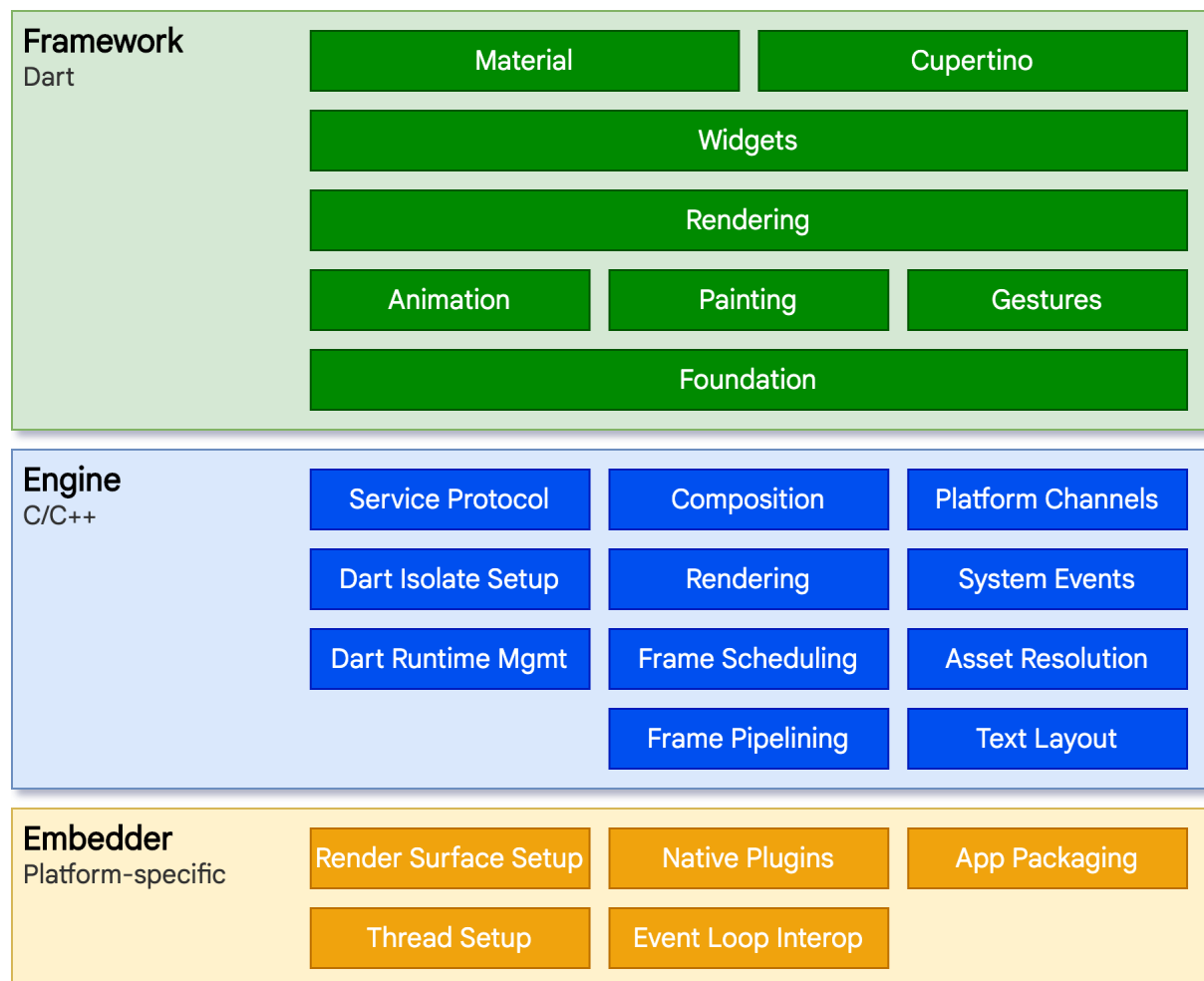


Figure 2: Architecture Of Flutter Framework

The operational workflow of the Jewellery Marketing and Saving Schemes App begins with the initiation of user interaction through the Flutter mobile application. A user first registers or logs into the system using secure authentication methods provided by the backend service. Once authenticated, the user can browse jewellery products, explore available saving schemes, and select a suitable plan based on their preferences. When a user enrolls in a saving scheme, the application records essential details such as user information, selected scheme, installment amount, and duration in the cloud database. Upon performing actions such as making an installment payment or purchasing jewellery, the transaction request is securely processed through an integrated digital payment system. The payment gateway verifies the transaction details, ensures sufficient balance, and completes the transaction using encrypted communication protocols. Once the payment is successful, the transaction details are stored in the database and updated in real time, allowing users to track their payment history and scheme progress instantly. The system continuously interacts with the backend services, such as Firebase, to manage data efficiently. All user activities, including registration, scheme enrollment, and transactions, are synchronized with the cloud database, ensuring consistency and accuracy. The application logic processes this data and dynamically updates the user interface to reflect the latest information. Notifications are generated for important events such as payment confirmations, due date reminders, and promotional offers, which are delivered to users through a notification service.

From an administrative perspective, the system allows authorized users (admins) to manage jewellery products, create and update saving schemes, and monitor customer activities. The admin panel communicates with the same backend infrastructure, ensuring centralized control and efficient data management. The system architecture ensures that all operations are performed in a structured and secure manner, minimizing errors and enhancing user experience. The application's workflow is designed to be efficient, user-friendly, and scalable. It ensures real-time data processing, secure transactions, and seamless interaction between the frontend and backend components. Key characteristics of the system include reliability, transparency in transaction tracking, and ease of access through mobile devices. By leveraging modern technologies such as Flutter, Dart, cloud databases, and digital payment systems, the Jewellery Marketing and Saving Schemes App provides a robust solution for managing jewellery investments and marketing activities. Despite its advantages, the system may face challenges such as network dependency, data security concerns, and integration complexities with third-party payment services. However, with proper implementation of security measures and scalable infrastructure, these challenges can be effectively managed.

2.Dart Programming Language:

In the context of blockchain technology, a block serves as the fundamental data unit that records a collection of digital transactions in a structured, verifiable, and immutable format. Each block contributes to a chronological sequence within the blockchain, forming a secure ledger that resists tampering and promotes decentralized consensus. A block functions both as a repository of data and as a structural component that ensures the integrity of the entire system through cryptographic linkages. By design, each block is cryptographically connected to its predecessor, enabling a linear and append-only data structure that prevents unauthorized alterations. This design underpins blockchain's immutability and trustless nature, making it suitable for decentralized applications across domains such as finance, supply chain, healthcare, and governance.

This figure represents the internal architecture and working of the Dart programming language, which plays a vital role in developing the Jewellery Marketing and Saving Schemes App. The architecture is designed to support high performance, fast development, and efficient execution across different platforms. At the core of the architecture is the Dart Virtual Machine (Dart VM), which is responsible for executing Dart code during the development phase. It supports Just In Time (JIT) compilation, allowing the code to be compiled and executed dynamically. This feature enables Flutter's hot reload functionality, which helps developers instantly apply code changes without restarting the application, significantly improving development speed and productivity. For production environments, Dart uses Ahead Of Time (AOT) compilation, where the Dart code is converted into native machine code before execution. This eliminates the need for runtime interpretation and results in faster startup time, improved performance, and better optimization. AOT compilation ensures that the final application runs smoothly on mobile devices with minimal latency. Another important component of the architecture is the core libraries, which provide essential functionalities such as data structures, file handling, asynchronous programming, and mathematical operations. These libraries simplify development by offering pre-built tools that developers can directly use in the application. In the Jewellery Marketing and Saving Schemes App, these libraries are used to manage user data, process transactions, and handle application logic efficiently. The architecture also includes a runtime environment, which manages memory allocation, garbage collection, and execution flow. The garbage collector automatically removes unused objects from memory, ensuring efficient resource utilization and preventing memory leaks. This is particularly important for maintaining the performance of mobile applications. Dart's support for asynchronous programming, through features like Futures and Streams, is another key aspect of its architecture.

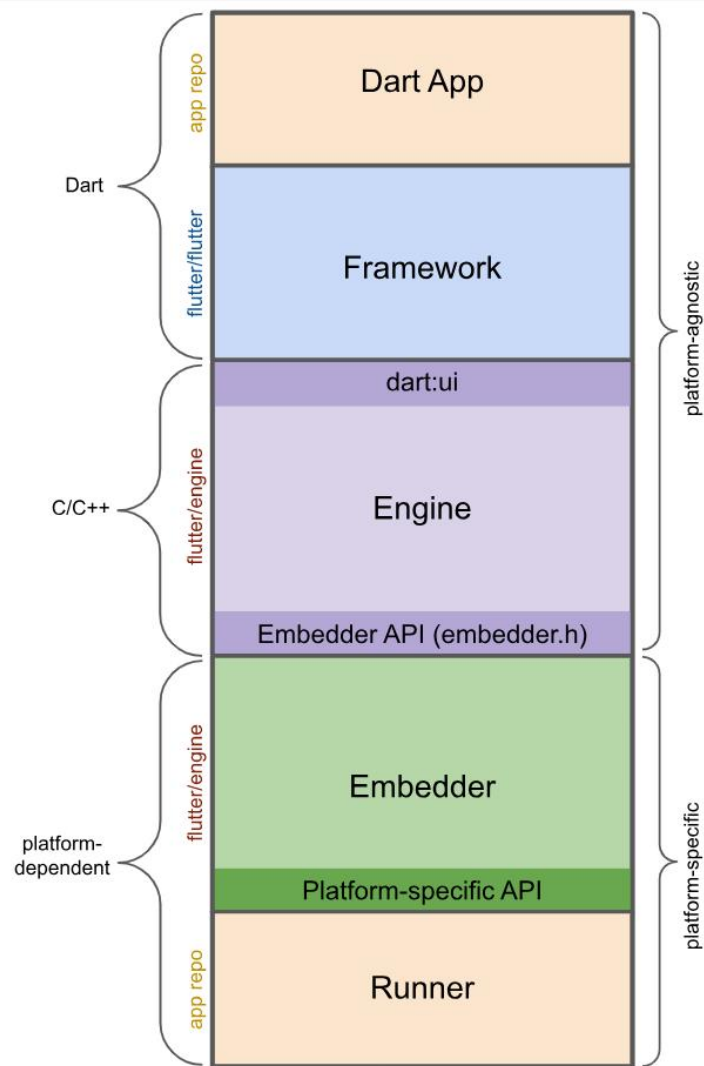


Figure 3: Architecture Of Dart Programming Language

3. Firebase Backend Service:

In the context of the Jewellery Marketing and Saving Schemes App, Firebase Backend Services act as the core infrastructure that supports data management, authentication, and real-time communication between the application and the cloud. Firebase is a comprehensive Backend-as-a-Service (BaaS) platform developed by Google, designed to simplify application development by providing scalable and secure backend solutions. It eliminates the need for managing traditional servers by offering ready-to-use services such as authentication, cloud database, storage, and notifications. Firebase enables seamless integration with Flutter applications, ensuring efficient handling of user data and system operations. The architecture of Firebase consists of multiple integrated services that work together to provide a complete backend solution. One of the primary components is Firebase Authentication, which ensures secure user registration and login using email, password, or third-party providers. It manages user identities and protects sensitive data from unauthorized access. Another key component is the Cloud Firestore Database, a NoSQL cloud database that stores application data such as user profiles, jewellery products, saving schemes, and transaction records. Firestore supports real-time data synchronization, allowing changes made in the database to be instantly reflected in the application. This ensures that users can view updated information such as payment status and scheme progress without delay. Firebase also provides Cloud Storage, which is used to store and manage media files such as jewellery images and documents. It ensures secure and efficient storage with controlled access permissions. Additionally, Firebase Cloud Messaging (FCM) is used to send push notifications to users, such as payment reminders, transaction confirmations, and promotional offers, thereby improving user engagement. The Firebase backend operates through a cloud-based infrastructure, where the Flutter application communicates with Firebase services via APIs. When a user performs an action, such as enrolling in a saving scheme or making a payment, the request is sent to Firebase, processed securely, and the result is updated in the database. This interaction ensures smooth and real-time communication between the frontend and backend. Firebase also includes built-in security features such as authentication rules and data access controls, which protect user information and ensure data integrity. Its scalable architecture allows the system to handle a growing number of users without affecting performance. In the context of the Jewellery Marketing and Saving Schemes App, this architecture enables the Flutter application to directly connect with Firebase for storing user data, managing saving schemes, processing transactions, and sending notifications. Firebase provides built-in security mechanisms such as authentication and access control rules, ensuring that user data is protected.

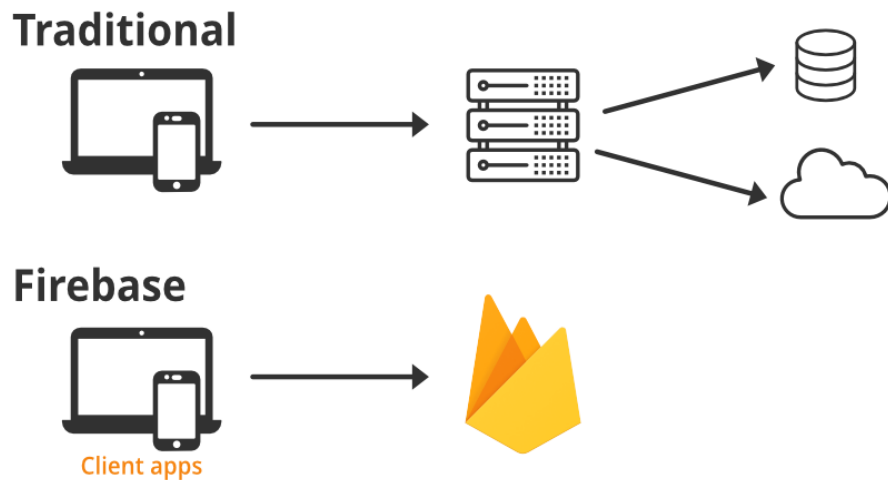


Figure 4: Firebase vs Traditional Database

This figure illustrates the comparison between the traditional backend architecture and the Firebase-based backend architecture used in modern mobile applications. In the traditional approach, client applications such as mobile or web apps communicate with a centralized server, which is responsible for processing requests, handling business logic, and interacting with databases and cloud storage. This model requires developers to manage server infrastructure, ensure scalability, handle security, and maintain system performance, which increases complexity and development time.

4.Digital Payment System:

A digital payment system is an electronic mechanism that facilitates the transfer of monetary value between users and service providers without the use of physical cash, ensuring secure, efficient, and real-time financial transactions. In the context of the Jewellery Marketing and Saving Schemes App, the digital payment system plays a critical role in enabling users to perform installment payments, enroll in saving schemes, and purchase jewellery products seamlessly. The system operates through secure communication protocols and integrates with trusted payment gateways to process transactions reliably. It functions as an automated financial module that manages payment initiation, authentication, processing, and confirmation, thereby eliminating the need for manual intervention and reducing the risk of errors. An illustrative example of the digital payment system within the application can be observed when a user enrolls in a monthly jewellery saving scheme. The user selects a scheme and initiates a payment using a preferred

method such as UPI, debit card, credit card, or mobile wallet. Once the payment request is initiated, the system securely transmits the transaction details to the payment gateway, which acts as an intermediary between the application and the banking network. The gateway encrypts sensitive information and forwards it to the respective bank for verification. The bank then validates the transaction by checking account balance, verifying credentials such as PIN or OTP, and ensuring compliance with security protocols. If the verification is successful, the transaction is approved and the payment amount is transferred to the merchant account. In case of failure, the transaction is declined, and the user is notified accordingly. The functional execution of the digital payment system begins with the user initiating a transaction through the application interface. The request is authenticated and securely transmitted to the payment gateway using encrypted communication channels. The gateway processes the request and interacts with the banking network to complete the transaction. Upon successful completion, a confirmation response is sent back to the application, and the transaction details are recorded in the cloud database. The system then updates the user interface in real time, reflecting the updated payment status and scheme progress. Notifications are also generated to inform the user about the transaction outcome, ensuring transparency and user engagement. To ensure security and reliability, the digital payment system incorporates multiple protection mechanisms such as encryption, tokenization, and multi-factor authentication. These techniques safeguard sensitive user data and prevent unauthorized access or fraudulent activities. Additionally, the system maintains a detailed record of all transactions, including timestamps, payment amounts, and transaction IDs, which can be used for auditing and tracking purposes. The integration of the digital payment system within the Jewellery Marketing and Saving Schemes App enhances user convenience by enabling fast and cashless transactions. It ensures transparency by providing real-time updates and transaction history, while also improving efficiency by automating financial operations. Despite its advantages, the system may face challenges such as network dependency, payment failures, and security threats; however, these issues can be mitigated through robust system design and the use of reliable payment gateways.

5.Cloud Database System:

A cloud database system is a centralized data storage and management solution hosted on remote servers, enabling real-time access, scalability, and secure handling of application data over the internet. In the context of the Jewellery Marketing and Saving Schemes App, the cloud database system plays a fundamental role in storing, organizing, and managing all critical information such as user details, jewellery products, saving schemes, and transaction records. Unlike traditional local

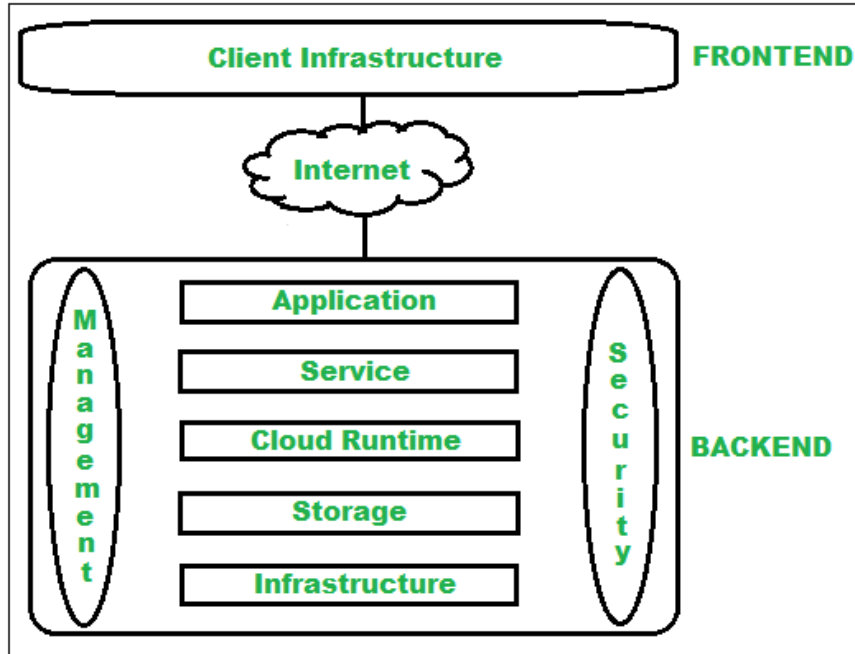


Figure 5: Architecture of Cloud Services

databases, cloud-based systems provide seamless data synchronization and accessibility from multiple devices, ensuring that users and administrators can interact with up-to-date information at any time. The system operates as a structured repository that maintains data consistency, integrity, and availability, thereby supporting the overall functionality of the application. An illustrative example of the cloud database system can be observed when a user registers and enrolls in a saving scheme. The user's personal information, selected scheme details, installment plan, and transaction history are securely stored in the cloud database. Whenever the user makes a payment or updates their profile, the changes are instantly reflected in the database and synchronized across the application. This ensures that users can track their savings progress and transaction history in real time without any delays or inconsistencies. To ensure reliability and performance, the cloud database system incorporates advanced features such as automatic data replica-

tion, backup, and recovery mechanisms. These features protect against data loss and ensure continuous availability even in the event of system failures. Security measures such as authentication, access control rules, and data encryption are implemented to safeguard sensitive user information and prevent unauthorized access. Additionally, the system is designed to handle large volumes of data and scale dynamically based on user demand, making it suitable for growing applications. The integration of a cloud database system within the Jewellery Marketing and Saving Schemes App significantly enhances efficiency and user experience. It eliminates the need for manual record-keeping, reduces errors, and provides a transparent and organized way to manage data. It ensures that the Jewellery Marketing and Saving Schemes App operates smoothly and reliably, providing users with a modern and convenient digital experience.

6.Notification System:

A notification system is an automated communication mechanism that delivers real-time alerts, updates, and reminders to users through digital channels such as mobile push notifications, SMS, or email. In the context of the Jewellery Marketing and Saving Schemes App, the notification system plays a crucial role in enhancing user engagement, ensuring timely actions, and maintaining effective communication between the application and its users. It functions as an intelligent alerting module that informs users about important events such as payment confirmations, installment due dates, scheme updates, and promotional offers. By providing timely notifications, the system reduces the chances of missed payments and improves overall user experience. An illustrative example of the notification system can be observed when a user enrolls in a saving scheme. After enrollment, the system automatically schedules reminders for upcoming installment payments based on the scheme duration. As the due date approaches, the user receives a push notification reminding them to complete the payment. Similarly, once a payment is successfully processed, a confirmation notification is sent to the user, ensuring transparency and trust. Promotional notifications may also be sent to inform users about new jewellery collections, discounts, or special offers. The functional execution of the notification system begins when a specific event is triggered within the application. These events may include user registration, payment completion, due date alerts, or administrative updates. Once an event occurs, the backend system processes the event and generates a notification message. This message is then delivered to the user's device through services such as Firebase Cloud Messaging (FCM). The system ensures that notifications are delivered efficiently and in real time, even when the application is not actively in use. To enhance effectiveness, the notification system supports personalization and scheduling features. Notifications can be customized based on user preferences, behavior, and activity

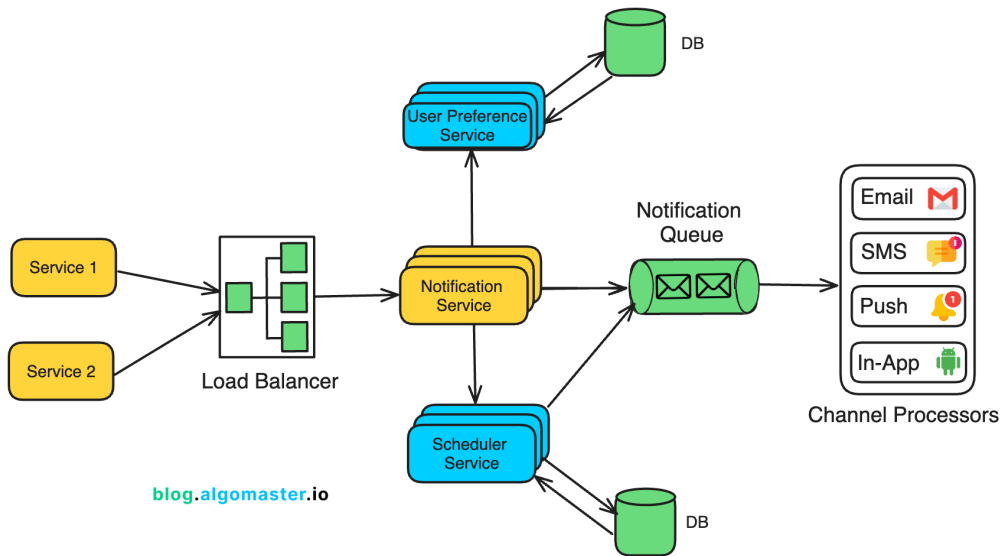


Figure 6: Notification System

within the application. For example, users who frequently engage with certain products may receive targeted promotional notifications. Additionally, scheduled notifications ensure that reminders are sent at appropriate times, improving user responsiveness and engagement. Security and reliability are important aspects of the notification system. The system ensures that only authorized messages are delivered to users and prevents unauthorized access or misuse. It also maintains logs of sent notifications, which can be used for monitoring and analysis. The use of cloud-based notification services ensures scalability, allowing the system to handle a large number of users simultaneously without performance issues. The integration of the notification system within the Jewellery Marketing and Saving Schemes App significantly enhances communication and user interaction. It ensures that users remain informed about their activities and responsibilities, thereby improving efficiency and satisfaction. Despite its advantages, challenges such as notification overload and dependency on internet connectivity may arise; however, these can be managed through proper configuration and user preference settings. Overall, the notification system serves as a vital component of the application, enabling real-time communication, improving user engagement, and ensuring smooth operation of saving schemes. It transforms the application into a proactive system that continuously interacts with users, providing timely and relevant information.

2.2 Platforms

1. Flutter SDK:



Figure 7: Flutter SDK

Flutter SDK is an open-source software development kit developed by Google, designed for building high-performance, cross-platform mobile applications using a single codebase. It provides a comprehensive framework for designing, developing, testing, and deploying applications for Android, iOS, web, and desktop platforms. Flutter SDK includes a rich set of pre-built widgets, development tools, and libraries that facilitate the creation of visually appealing and responsive user interfaces. Its widget-based architecture allows developers to construct complex UI components by combining smaller reusable elements, ensuring modularity and scalability in application design. From a technical perspective, Flutter SDK integrates seamlessly with the Dart programming language and utilizes the Skia rendering engine to draw UI components directly onto the screen. This eliminates the dependency on native UI components and ensures consistent performance across different platforms. The SDK also provides features such as hot reload, which allows developers to instantly apply code changes without restarting the application, significantly improving development efficiency.

2.Android Studio/ VS Studio:



Figure 8: Android Studio / VS Studio

Android Studio and Visual Studio Code are integrated development environments (IDEs) used for developing Flutter applications. Android Studio, developed by Google, provides a powerful environment with advanced features such as code editing, debugging, performance analysis, and emulator support. It includes built-in tools for managing project structure, running applications on virtual devices, and testing application performance. One of the key components of Android Studio is the Android Emulator, which allows developers to simulate different Android devices and test applications in a virtual environment. Visual Studio Code (VS Code), on the other hand, is a lightweight and highly customizable code editor developed by Microsoft. It supports Flutter development through extensions that provide features such as syntax highlighting, code completion, debugging, and version control integration. Both IDEs support Dart programming and offer seamless integration with Flutter SDK, enabling efficient application development. Overall, Android Studio serves as a powerful and comprehensive development platform that enhances the efficiency, reliability, and scalability of mobile application development. Its advanced tools, seamless integration with Flutter and Firebase, and robust debugging capabilities make it an essential platform for developing and maintaining the Jewellery Marketing and Saving Schemes App.

3. Firebase Console:



Figure 9: Firebase Console

Firebase Console is a cloud-based management platform provided by Google that serves as a centralized interface for configuring, monitoring, and managing back-end services used in mobile and web applications. It acts as the control panel for Firebase services, enabling developers to integrate essential backend functionalities such as authentication, real-time databases, cloud storage, analytics, and push notifications without the need to manage dedicated server infrastructure. In the context of the Jewellery Marketing and Saving Schemes App, Firebase Console plays a crucial role in handling application data, user management, and system operations efficiently and securely. From a technical standpoint, Firebase Console provides access to a wide range of services, including Firebase Authentication, Cloud Firestore Database, Cloud Storage, and Firebase Cloud Messaging (FCM). Firebase Authentication allows developers to implement secure user login and registration mechanisms using email, password, or third-party providers. Cloud Firestore is a NoSQL database that stores structured data such as user profiles, jewellery product details, saving scheme information, and transaction records. It supports real-time synchronization, ensuring that any changes in the database are instantly reflected in the application interface. The console also includes Cloud Storage, which is used for storing and managing media files such as jewellery images and related assets. This storage system ensures secure access and efficient retrieval of files.

2.3 Algorithms

1. Data Processing and Transaction Management Algorithms

In the context of the Jewellery Marketing and Saving Schemes App, data processing and transaction management algorithms play a crucial role in ensuring efficient handling of user data, scheme operations, and financial transactions. Unlike blockchain systems that rely on decentralized consensus mechanisms, this application operates on a centralized cloud-based architecture where algorithms are designed to manage data consistency, real-time updates, and secure transaction processing. These algorithms ensure that user actions such as registration, scheme enrollment, installment payments, and product purchases are executed accurately and efficiently. The system utilizes structured data handling algorithms to store and retrieve information from the cloud database. When a user performs an action, the request is processed through backend services, validated, and updated in the database. Transaction management algorithms ensure that financial operations such as payments are executed securely and consistently, preventing issues such as duplicate transactions or data inconsistencies. Additionally, real-time synchronization algorithms enable instant updates across the application, ensuring that users always have access to the latest information regarding their savings and transactions. The system also incorporates authentication and validation algorithms to verify user identity and ensure secure access to application features. These algorithms check user credentials, validate inputs, and enforce access control rules. Notification scheduling algorithms are used to trigger reminders and alerts based on predefined conditions such as due dates or successful transactions. Overall, these algorithms ensure reliability, efficiency, and security in the operation of the Jewellery Marketing and Saving Schemes App.

Algorithm Name	Use	Challenges in Implementation	Complexity in Implementation
Authentication Algorithm	Used for verifying user identity during login and registration.	Ensuring data security and preventing unauthorized access.	Medium
Transaction Processing Algorithm	Used for handling payments and updating transaction records securely.	Handling payment failures, ensuring data consistency.	High

CHAPTER 2. LITERATURE REVIEW

Algorithm Name	Use	Challenges in Implementation	Complexity in Implementation
Data Synchronization Algorithm	Used for real-time updates between app and cloud database.	Network dependency and latency issues.	Medium
Recommendation Algorithm	Used to suggest jewellery products based on user preferences.	Requires accurate user data and personalization logic.	Medium
Notification Scheduling Algorithm	Used to send reminders for payments and updates.	Avoiding notification overload and ensuring timely delivery.	Low- Medium

CHAPTER 3: Design, Development and Drawing

3.1 Proposed Work

3.1.1 Problem Statement

Traditional jewellery saving schemes and product browsing are often managed manually or through basic systems, making it difficult for customers to track installments, view scheme benefits, or browse jewellery collections conveniently. This leads to confusion, lack of transparency, and reduced customer engagement. To address these challenges, a mobile application will be developed that allows users to explore jewellery products, enroll in saving schemes, track payments, and receive reminders—all through a secure, user-friendly, and interactive digital platform.

3.1.2 Objectives

1. To develop a mobile application that enables customers to browse jewellery collections digitally.
2. To implement a saving scheme module allowing users to enroll, view installment schedules, and track payments.
3. To provide an admin dashboard for managing jewellery items, scheme plans, and customer records efficiently.

3.2 Design

3.2.1 System architecture

The system architecture of the Jewellery Marketing and Saving Schemes Application defines how different components communicate to provide a smooth and secure user experience. The workflow begins with the customer interacting with the mobile app interface, where they browse jewellery items, view saving schemes,

and manage their payments. All user requests pass through the app UI, which captures actions and forwards them to the underlying processing modules.

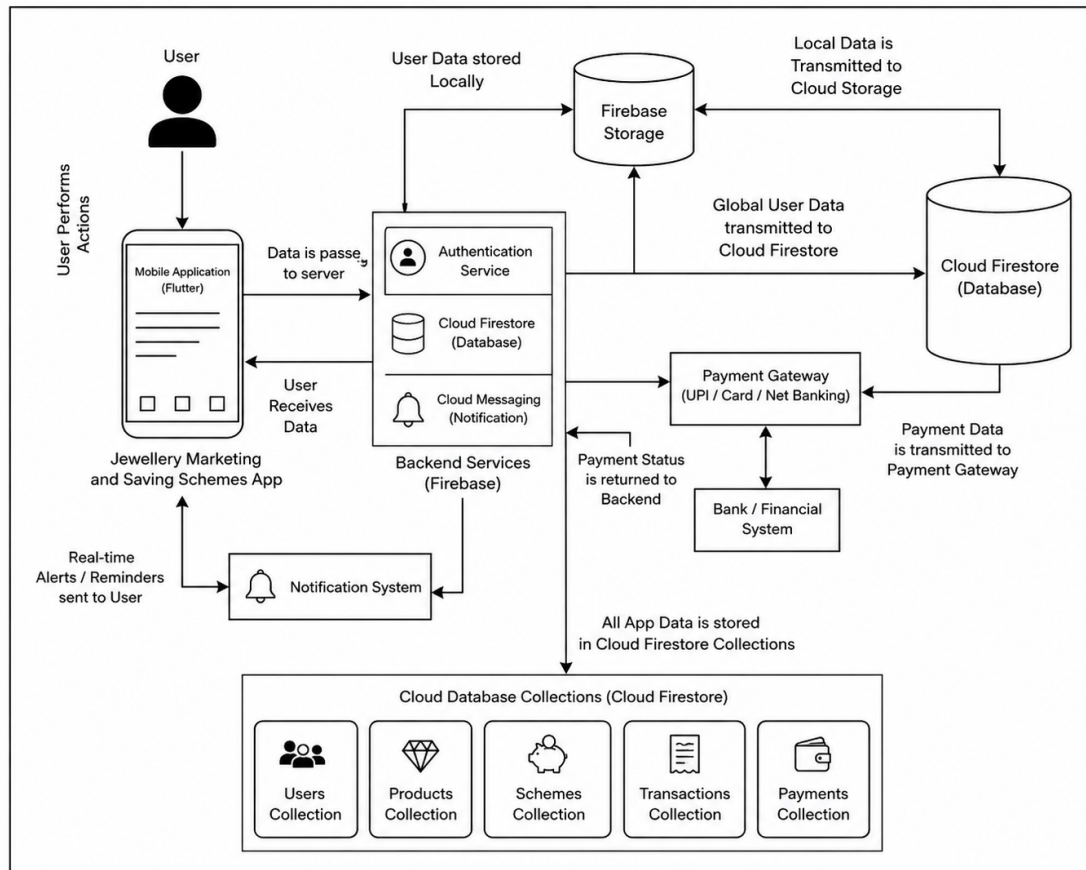


Figure 10: System Architecture of Jewellery Marketing and Saving Schemes App

Figure 10: System Architecture

Phase I: User Interaction, Authentication, and Data Initialization

The first phase of the Jewellery Marketing and Saving Schemes App begins with user interaction through the mobile application interface. The user performs actions such as registration, login, browsing jewellery products, and selecting saving schemes. These inputs are captured by the Flutter-based mobile application and transmitted to the backend system. The application sends the user data to Firebase backend services, where authentication is performed to verify the identity of the user. Once authenticated, the data is processed and stored in the cloud database (Cloud Firestore). This phase ensures secure user access, proper data validation, and structured storage of user information. It forms the initial stage

of the system workflow by establishing communication between the user and the backend infrastructure. The validated data is then stored in the Cloud Firestore database, where it is organized into structured collections. This phase establishes a secure communication channel between the frontend and backend while ensuring accurate data entry and initialization. It forms the foundational layer of the system by enabling reliable user access and preparing the system for further operations.

Phase II: Business Logic Processing and Scheme Management

The second phase focuses on processing user requests and executing the core business logic of the application. When a user interacts with the system—such as enrolling in a saving scheme, updating preferences, or browsing product details—the request is processed by the backend services. Firebase handles these operations by interacting with the Cloud Firestore database, retrieving relevant data, and applying application logic. The system calculates scheme-related parameters such as installment amount, duration, and eligibility conditions. It ensures that all operations follow predefined rules and constraints, thereby maintaining system consistency and reliability. This phase also includes real-time data synchronization, where any changes made in the database are instantly reflected in the application interface. The use of asynchronous programming ensures that these operations are performed efficiently without affecting application performance. This phase acts as the core computational layer, where all application logic and decision-making processes are executed. A key aspect of this phase is real-time data synchronization, enabled by Firebase Firestore. Any updates made in the database—such as changes in user profile, scheme enrollment status, or transaction records—are instantly reflected across all connected devices. This eliminates the need for manual data refresh and ensures that users always have access to the most recent information. The system also leverages asynchronous programming models to handle multiple requests simultaneously without blocking execution. Operations such as database queries, API calls, and background processing are executed in a non-blocking manner, ensuring smooth application performance even under high user load. This improves responsiveness and enhances the overall user experience. Furthermore, this phase incorporates error handling and exception management mechanisms to deal with unexpected scenarios such as invalid inputs, network failures, or database inconsistencies. Appropriate fallback actions and user notifications are triggered to ensure system reliability. From a structural perspective, this phase follows a modular approach, where different functionalities such as scheme management, user data processing, and transaction preparation are handled by separate logical components. This modularity improves maintainability and allows easy updates or enhancements to the system without affecting overall functionality.

Phase III: Data Storage, Notification, and Response Delivery

The final phase of the Jewellery Marketing and Saving Schemes App focuses on persistent data storage, real-time communication, and delivery of processed responses to the user. Once the backend system completes business logic execution and transaction processing, all resulting data is securely stored in the cloud database infrastructure. This phase ensures that system outputs are properly recorded, synchronized, and communicated to users in an efficient and reliable manner. At the core of this phase is the Cloud Firestore database, which organizes application data into structured collections such as Users, Products, Schemes, Transactions, and Payments. Each data entity is stored as a document with unique identifiers, timestamps, and associated metadata. This structured approach enables efficient querying, indexing, and retrieval of data. The system ensures data consistency by maintaining atomic operations and validating write requests before committing them to the database. A key feature of this phase is real-time data synchronization, which allows any updates in the database to be instantly reflected across all connected client devices. Firebase's real-time capabilities ensure that users can view updated information such as payment status, scheme progress, and transaction history without manual refresh operations. This significantly enhances user experience and ensures transparency within the system. The notification subsystem operates in parallel to the data storage process. It follows an event-driven architecture, where specific system events—such as successful payments, failed transactions, upcoming due dates, or promotional updates—trigger the generation of notifications. These notifications are processed and delivered to users through Firebase Cloud Messaging (FCM). The system supports multiple notification types, including transactional alerts, reminder notifications, and promotional messages, ensuring targeted and relevant communication. Additionally, the system incorporates response handling mechanisms that transmit processed results back to the frontend application. Once data is stored and validated, the backend sends structured responses to the Flutter application, enabling users to view updated details such as account status, scheme enrollment confirmation, and payment receipts. This response delivery follows a secure request-response model combined with real-time streaming for continuous updates.

3.3 Development

3.3.1 Software Requirements Specifications

1. Software Compatibility

- Operating Systems:
 - Windows 10/11
 - macOS
 - Linux (Ubuntu 20.04 or later)
 - Android (for mobile application testing)
 - iOS (optional for cross-platform support)

2. Programming Languages

- Dart (Flutter application development)
- JavaScript (optional for backend integration)
- HTML, CSS (for web dashboard if required)

3. Development Tools

- Android Studio
- Visual Studio Code
- Flutter SDK
- Firebase Console
- Git and GitHub
- Android Emulator / Physical Device

4. Libraries and Frameworks

- **Flutter Framework:** Cross-platform mobile application development.
- **Firebase SDK:** Backend services including authentication, database, and notifications.
- **Provider / Riverpod / Bloc:** State management in Flutter.
- **HTTP / Dio:** API communication.

5. Backend and Deployment

- **Backend Platform:** Firebase (BaaS)
- **Database:** Cloud Firestore (NoSQL Database)
- **Storage:** Firebase Cloud Storage
- **Authentication:** Firebase Authentication
- **Hosting and Deployment:** Firebase Hosting / GitHub

3.3.2 Hardware Requirements Specifications

1. Server-Side Requirements

- **Minimum Configuration:**
 - Cloud-based server (Firebase)
 - Stable internet connection
- **Recommended Configuration:**
 - Scalable cloud infrastructure (Firebase Services)
 - High availability and redundancy

2. Client-Side Requirements

- Hardware:
 - Android Smartphone (minimum 4 GB RAM)
 - Modern OS (Android 8.0 or above)
- Software:
 - Mobile application installed
 - Internet connectivity for real-time operations

3. Development Environment

- Hardware:
 - Quad-core processor
 - 8 GB RAM (minimum)
 - 256 GB SSD
- Required:
 - High-speed Internet connection
 - Android Emulator or physical testing device

3.4 Drawing

3.4.1 Data Flow Diagram Level 0

The Level 0 Data Flow Diagram for the Jewellery Marketing and Saving Schemes Application provides a high-level overview of how the system interacts with its external entities, including customers, admin/jewellers, the payment gateway, and the application database. Customers send browsing queries, explore jewellery products, and join saving schemes, all of which are processed by the central Jewellery Marketing and Saving Schemes System. Admins or jewellers manage product details and configure saving schemes, which the system updates and stores in the database. The system retrieves required data—such as user details, scheme information, and payment records—from the App Database whenever needed. When a customer makes a payment, the system sends a payment request to the Payment Gateway, receives the payment status, and updates the user’s scheme progress accordingly. Additionally, the system interacts with the Marketing Offers module to generate personalized recommendations and promotional information. Overall, the context diagram highlights how the system serves as the central hub, coordinating communication between users, administrators, database operations, and external payment services.



Figure 11: Data Flow Diagram level 0

3.4.2 Data Flow Diagram Level 1

The Level 1 Data Flow Diagram for the Jewellery Marketing and Saving Schemes Application illustrates the detailed internal operations of the system and how it processes interactions from both customers and the admin/jeweller. Customers begin by registering or logging into the application, after which they can browse jewellery products, view catalog details, and explore saving schemes. These actions are handled by the Product and Catalog Management module, which retrieves or stores product information in the App Database. When customers join a saving scheme or check scheme details, the Saving Scheme Management module processes their request by accessing scheme records stored in the database. The User Management module handles profile data and ensures proper authentication and access control. Overall, the Level 1 DFD shows how the system coordinates product management, scheme processing, user interactions, and payment operations to provide a seamless and efficient experience.

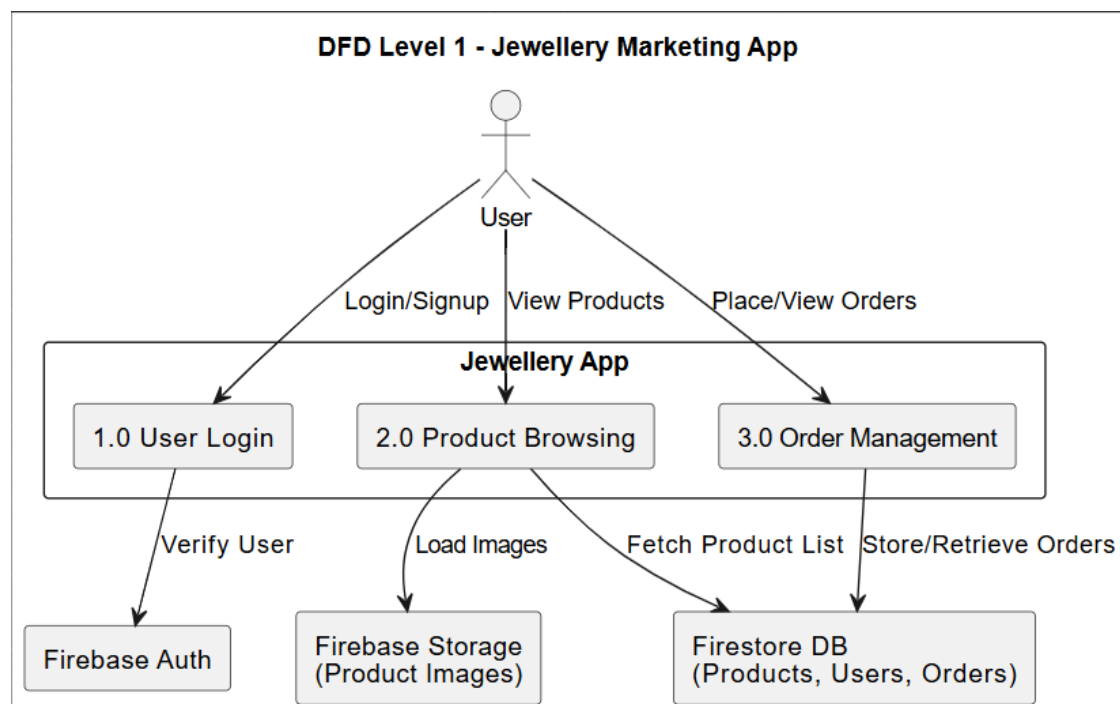


Figure 12: Data Flow Diagram level 1

3.4.3 Data Flow Diagram Level 2

The Level 2 Data Flow Diagram provides a detailed view of how the Jewellery Marketing and Saving Schemes App processes user and admin actions. When customers browse the catalog or search for products, the system retrieves relevant jewellery data from the App Database and returns it through the Catalog Management module. For saving schemes, the Scheme Management process handles enrollment, fetches scheme details, and stores new entries in the database. Payment requests initiated by users flow through the Payment Processing unit, which communicates with the Payment Gateway and updates payment records after confirmation. Admins manage products and schemes through dedicated processes that store or update information in the database. Additionally, the system triggers promotional alerts and reminders through the Notifications module whenever required. Overall, Level 2 highlights how catalog browsing, scheme enrollment, payments, admin actions, and notifications are executed and synchronized with the central database.

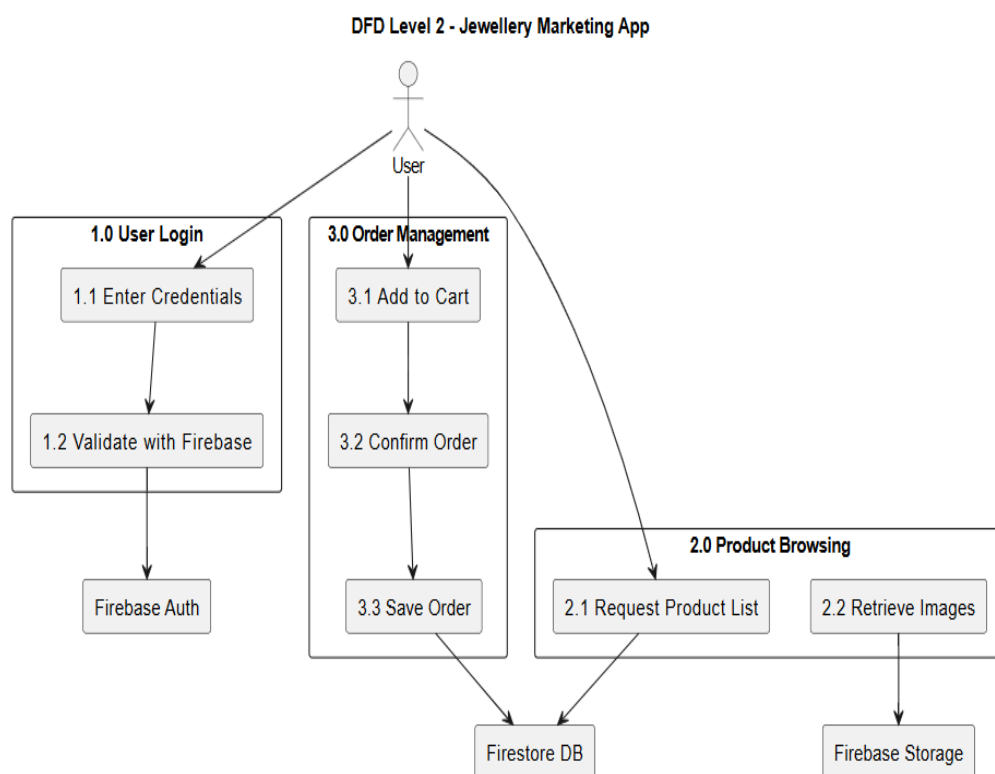


Figure 13: Data Flow Diagram level 2

3.5 UML Diagram

3.5.1 Use Case Diagram

The Use Case Diagram for the Jewellery Marketing and Saving Schemes Application illustrates the interaction between three main actors: the Customer, the Admin/Jeweller, and the Payment Gateway. Customers can register or log in, browse the jewellery catalog, view product details, join saving schemes, check their scheme status, receive personalized offers, and make payments for purchases or scheme top-ups. The Payment Gateway processes these payments and returns the payment status to the system. The Admin/Jeweller manages the backend operations by adding or updating products, configuring saving schemes, and viewing customer enrollments. Together, these use cases represent the core functionality of the system, enabling smooth browsing, secure transactions, and efficient scheme management within the application.

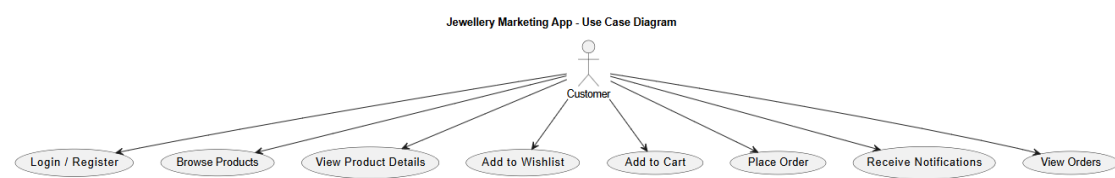


Figure 14: Use Case Diagram

3.5.2 Sequence Diagram

The Sequence Diagram for the Jewellery Marketing and Saving Schemes Application illustrates how different system components interact to support product browsing, scheme enrollment, and payment processing. The customer begins by accessing the Mobile/Web UI to browse jewellery items, after which the Product Service retrieves the required catalog data from the App Database and returns it to the user. When the customer selects a product or saving scheme, the Scheme Service fetches detailed information and displays it through the interface. If the user chooses to join a saving scheme, an enrollment request is sent to the Scheme Service, which creates and stores the enrollment record in the database. During payment, the Payment Service generates a payment intent and communicates with the external Payment Gateway to process the transaction. Based on the payment result, the system updates the user's scheme status, records the transaction, and triggers the Notification Service to send an email or SMS receipt.

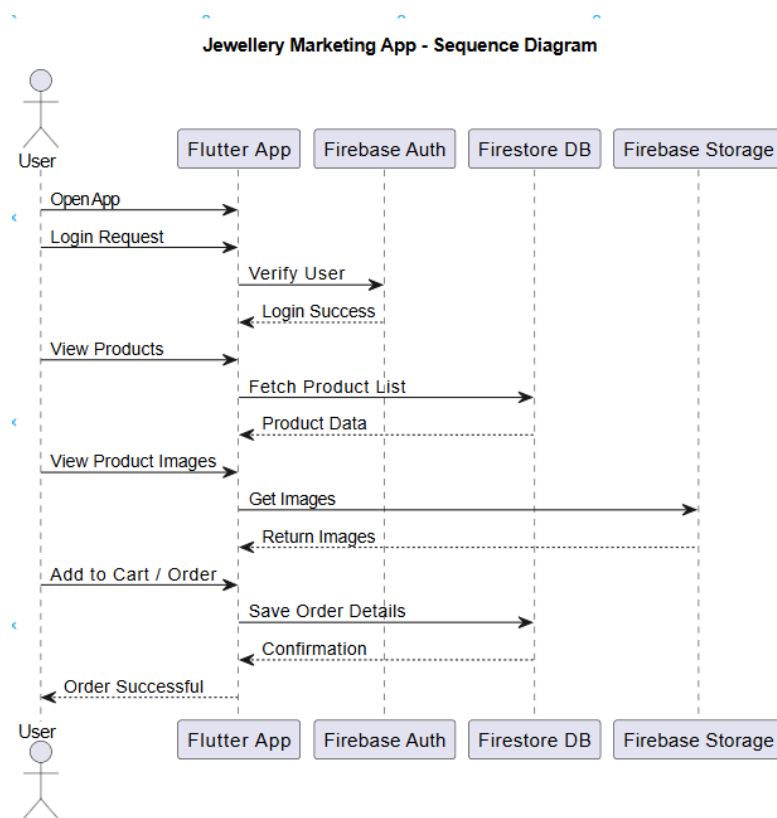


Figure 15: Sequence Diagram

3.5.3 Deployment Diagram

The Deployment Diagram illustrates how the Jewellery Marketing and Saving Schemes App operates across different system environments. Users access the mobile/web app on their devices, which send secure API requests to the Application Server. The server hosts key services—Product and Catalog Service, Saving Scheme Service, Payment Service, and Recommendation/Chatbot Service—that handle user actions and business logic. These services communicate with the Database Server to store and retrieve user data, jewellery information, scheme details, and transactions. For payments, the Payment Service connects to an external Payment Gateway to process transactions securely. This deployment setup ensures smooth user interaction, reliable processing, and secure data management.

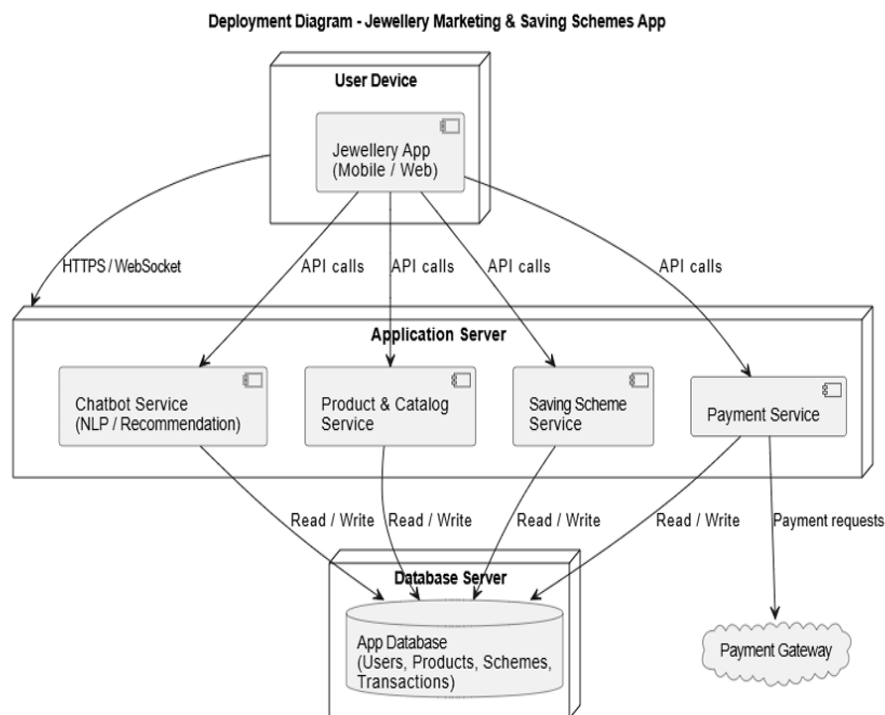


Figure 16: Deployment Diagram

3.5.4 Activity Diagram

The Activity Diagram for the Jewellery Marketing and Saving Schemes App describes the user's workflow when exploring products or joining a saving scheme. The process begins when the user opens the app and browses the jewellery catalog. After selecting a product, the system displays detailed information and identifies whether the user intends to simply view details or join a saving scheme. If the user chooses to join a scheme, the app fetches relevant scheme details and prepares the necessary response. Depending on the scheme requirements, the user may need to initiate a payment, after which the system verifies the transaction status. If no payment is needed, the user proceeds directly. Finally, the system compiles the appropriate response—either product information or scheme enrollment details—and delivers it back to the user, completing the activity flow.

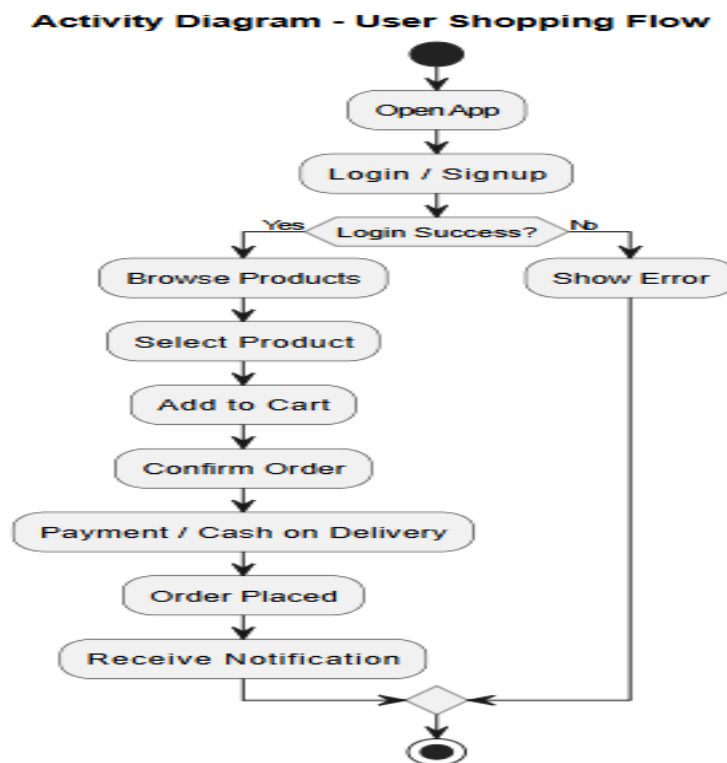


Figure 17: Activity Diagram

3.5.5 Class Diagram

The Class Diagram illustrates the main components of the Jewellery Marketing and Saving Schemes App and their roles. The User class represents customers who browse products and view scheme information. The Chatbot class processes user messages and identifies their intent. The Product class manages jewellery details, while the SavingScheme class handles scheme enrollment and tracking. The Payment class processes transactions and checks payment status. Together, these classes define the system's structure, enabling smooth interaction between users, products, schemes, and payment functions.

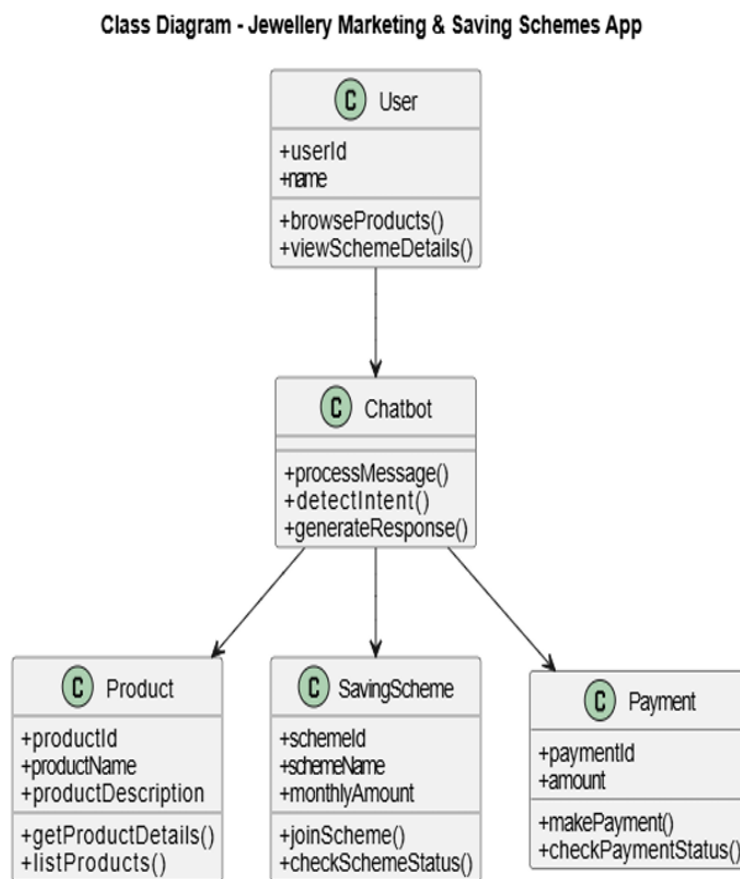


Figure 18: Class Diagram

3.5.6 Object Diagram

The Object Diagram illustrates a real example of how different instances in the Jewellery Marketing and Saving Schemes App interact. It shows a specific user (User1) communicating with the chatbot (Chatbot1), which identifies the intent to join a saving scheme. The user views a jewellery item represented by Product1, then proceeds to enroll in a plan captured by SavingEnrollment1. A related transaction is recorded through Payment1. Together, these objects represent the flow of a user selecting a product, joining a scheme, and completing a payment.

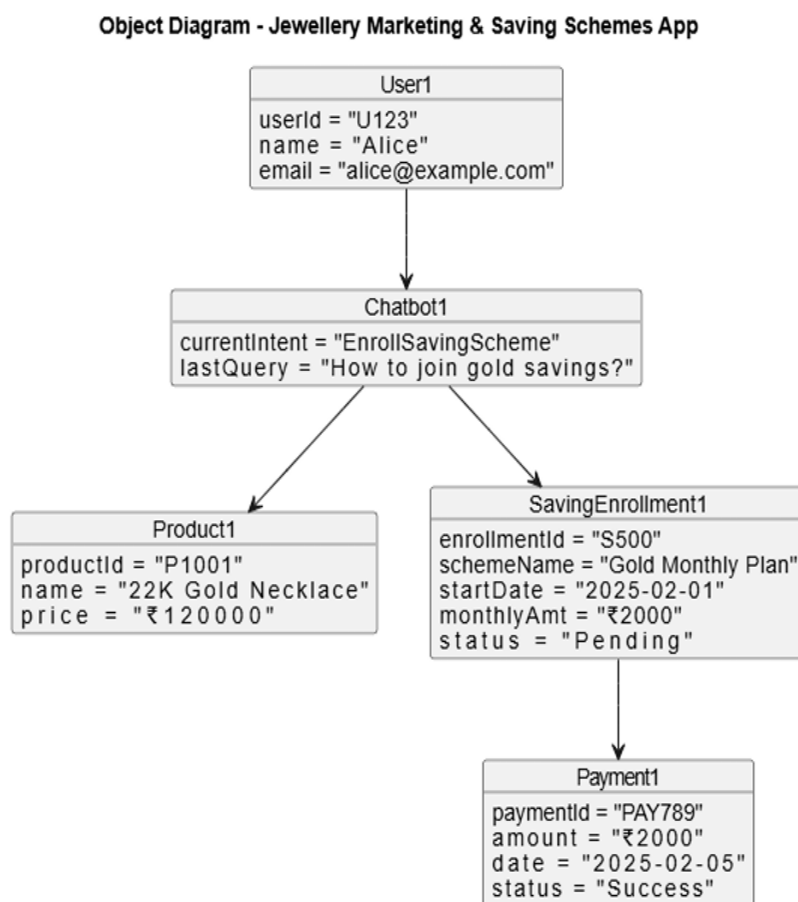


Figure 19: Object Diagram

3.6 Timeline of Project

Activity	Week 1	Week 2	Week 3-4	Week 5-6	Week 7-15	Week 16-32	Week 33-34	Week 35-39	Week 40
Project Stage I:									
1. Group Formation									
2. Topic Finalization									
3. Literature Survey / Info Collection									
4. Technical Specifications									
5. Synopsis									
6. start Block Diagrams, Design, Finalization of Drawings, Simulation/ Analysis, Process Sheet									
Project Stage II:									
7. Start of Experimentation / Procurement, Manufacturing, Assembly, Working Model, Experimentation, Implementation of final System, Parameter Measurement									
8. Publishing Research Paper (Minimum One paper in UGC/ Scopus indexed Journals/Conference)									
9. Exhibition & Presentation									
10. Report Completion									

Figure 20: Timeline of Project

CHAPTER 4. Experimentation, Results and Discussion

4.1 Mathematical Model

Basic Set Definition

- $U = \{u_1, u_2, u_3, \dots\}$: Set of users
- $P = \{p_1, p_2, p_3, \dots\}$: Set of jewellery products
- $C \subseteq P$: Cart (selected products)
- G : Current gold rate (per gram)
- W : Amount invested in digital gold
- L : Selected language (English, Hindi, Marathi)

Functions

1. **Authentication Function:**

$$\text{Auth}(U) \rightarrow \text{Access Granted} / \text{Denied}$$

2. **Product Selection Function:**

$$\text{Select}(U, P) \rightarrow C$$

3. **Cart Total Calculation:**

$$\text{Total} = \sum_{i=1}^n p_i$$

4. **Digital Gold Calculation:**

$$\text{Gold} = \frac{W}{G}$$

5. Language Selection Function:

$$\text{Lang}(U) \rightarrow L$$

Output

The system provides the following outputs:

- Display of selected jewellery
- Total cart value
- Gold quantity purchased
- Content in selected language

This model ensures a structured flow of data and functionality within the application.

4.2 Testing and Evaluation

1 Testing

Testing is an important phase to ensure that the application works correctly and efficiently. The Gemzi application was tested using different testing methods:

1. Unit Testing

Each module such as login, product display, cart, wallet, and AR try-on was tested individually to ensure correct functionality.

2. Integration Testing

All modules were combined and tested together to verify smooth interaction between components like frontend, backend, and database.

3. System Testing

The complete system was tested as a whole to ensure that all features work properly in real-time conditions.

4. User Interface Testing

The UI was tested to ensure that all buttons, navigation, and layouts are user-friendly and responsive.

5. Functional Testing

Key functionalities such as login, add to cart, gold purchase, and language selection were tested for correct outputs.

6. Performance Testing

The app was tested for speed, responsiveness, and smooth performance under different conditions.

7. Security Testing

Authentication and user data handling were tested to ensure safe and secure access.

2 Evaluation

The evaluation of the Gemzi Jewellery Application was done based on performance, usability, and functionality.

- The application provides a smooth and user-friendly interface, making it easy for users to navigate.
- Features like AR Try-On and Digital Gold Wallet enhance user experience and make the app innovative.
- The multi-language support (English, Hindi, Marathi) improves accessibility for different users.
- The Admin Dashboard allows efficient management of products and users.
- The system performs efficiently with minimal errors and good response time.

CHAPTER 4. Experimentation, Results and Discussion

Overall Evaluation

The application successfully meets its objectives by providing a reliable, interactive, and modern jewellery shopping platform. It combines technology and usability effectively, making it beneficial for both users and administrators.

No.	Test Category	Input / Action	Expected Output	Result
1	Login (Functional)	Enter valid email and password	User successfully logged in	Pass
2	Input Validation (Negative)	Leave login fields empty	Error message displayed	Pass
3	Product Selection	Select jewellery item	Product added to cart	Pass
4	Add to Cart (Functional)	Click “Add” button	Item added to cart list	Pass
5	AR Try-On (Feature)	Open camera and select earrings	Jewellery displayed on user face	Pass
6	Digital Gold Wallet	Enter amount (500)	Gold grams calculated correctly	Pass
7	Live Gold Rate (Data)	Open gold rate page	Updated 22K and 24K prices shown	Pass
8	Language Translation	Change language to Hindi/Marathi	App content translated	Pass
9	User Interface (UI)	Navigate between screens	Smooth and responsive UI	Pass
10	Admin Product Management	Add new product from dashboard	Product added and displayed	Pass

4.3 Project Screenshot

1. Welcome Screen

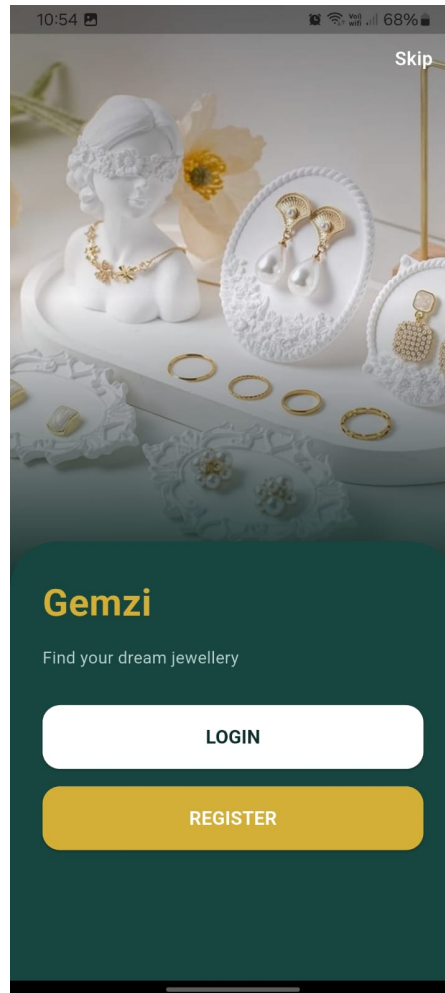


Figure 21: Welcome Screen

This screen is the first interface users see when they open the app. It introduces the app (Gemzi) and provides options to Login or Register, allowing users to access or create their account.

2. Registration Page

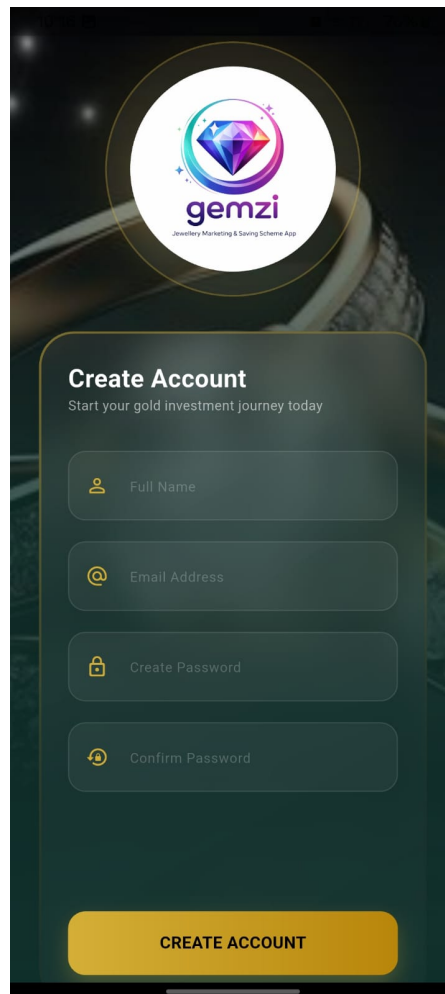


Figure 22: Registration Page

The Registration Page is designed for new users to create an account in the application. Users need to provide necessary details such as name, email, phone number, and password to register successfully.

3. Login Page

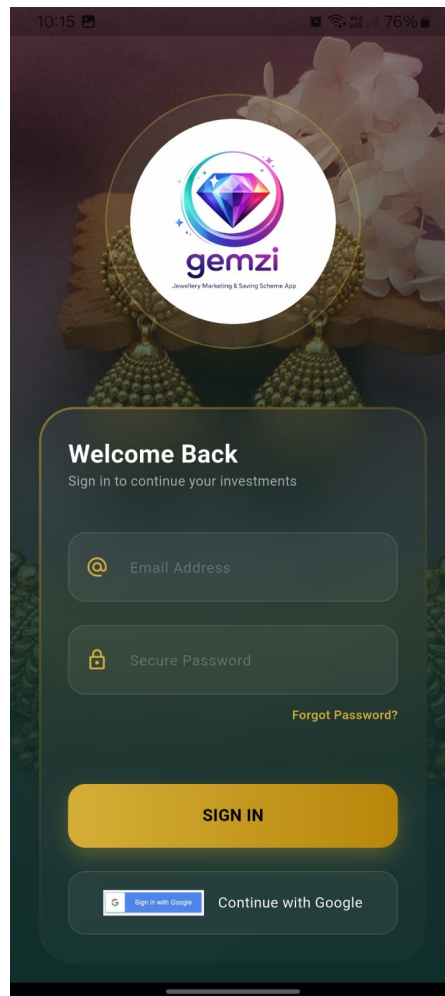


Figure 23: Login Page

The Login Page allows existing users to securely access their account in the application. Users are required to enter their registered credentials such as email and password. It also includes features like “Forgot Password” for password recovery and validation messages for incorrect inputs. After successful authentication, the user is redirected to the main dashboard or home screen.

4. Home Screen

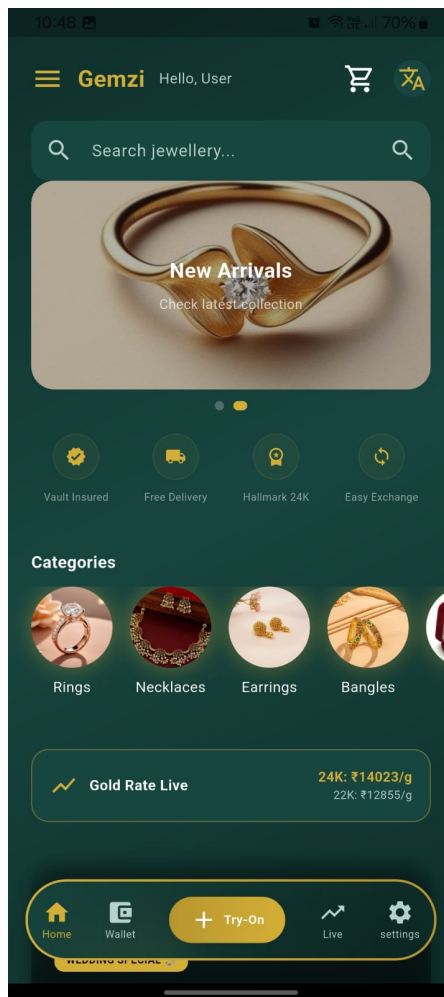


Figure 24: Home Screen

The Home Screen is the main interface of the Gemzi application, designed to provide users with a smooth and engaging shopping experience. It acts as the central hub where users can explore jewellery collections, view offers, and access key features of the app.

5. Exclusive Collection page

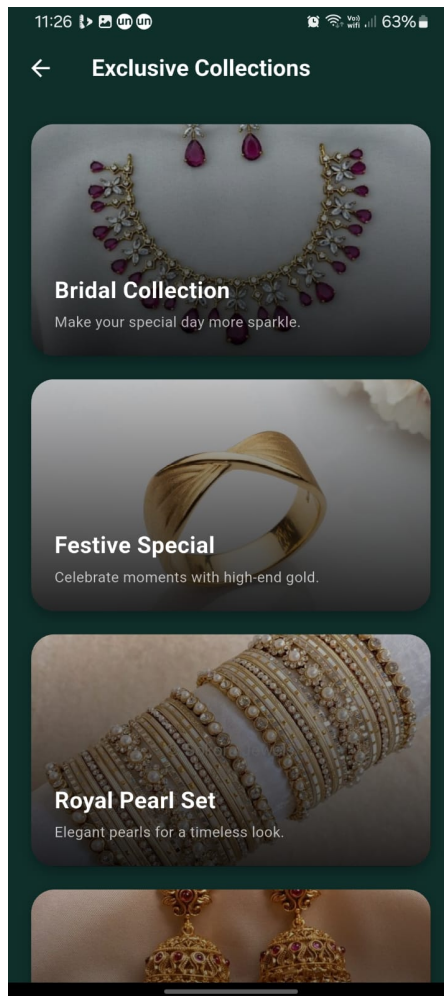


Figure 25: Exclusive Collection page

The Exclusive Collection Page displays specially curated jewellery categories such as Bridal Collection, Festive Special, Royal Pearl Set, etc. Each section showcases themed designs to help users easily explore premium and occasion-based jewellery. This page enhances user experience by organizing products into attractive and meaningful collections.

6. Wallet Page

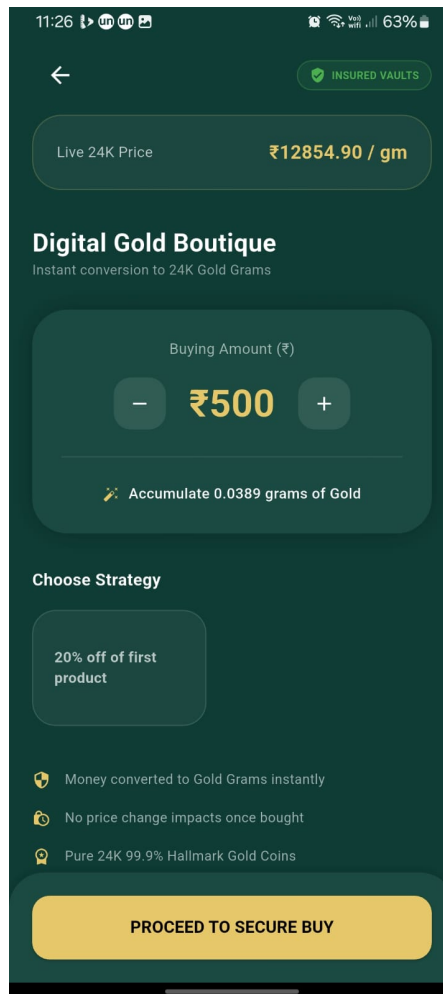


Figure 26: Wallet Page

The Wallet Page allows users to invest in digital gold. It shows the live 24K gold price and lets users enter an amount to buy gold in grams. The page also highlights benefits like secure storage, instant conversion, and price protection. It provides a simple and safe way for users to purchase and manage digital gold.

7. AR-Try On Page

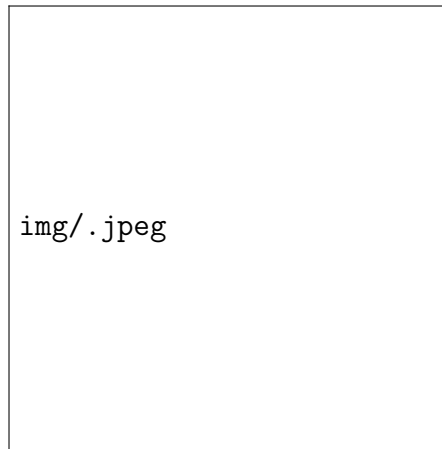


Figure 27: AR-Try On Page

The AR Try-On Page allows users to virtually try jewellery using their device camera. This feature uses Augmented Reality (AR) to display jewellery items like earrings, necklaces, rings, and bangles on the user in real-time.

8. Live-Rate Page

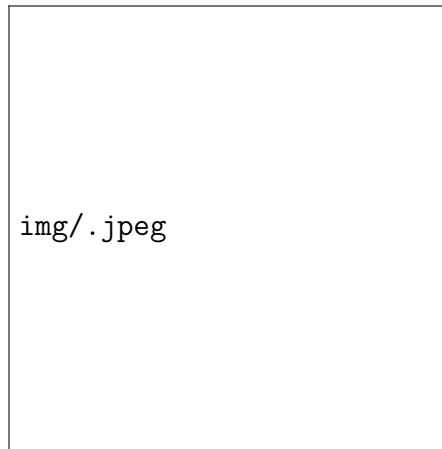


Figure 28: Live-Rate Page

The Live Gold Rate Page displays updated gold prices for 24K and 22K gold along with recent trends. It helps users track daily price changes and make informed buying decisions. The page ensures transparency by showing real-time data and market trends.

9. Setting Page

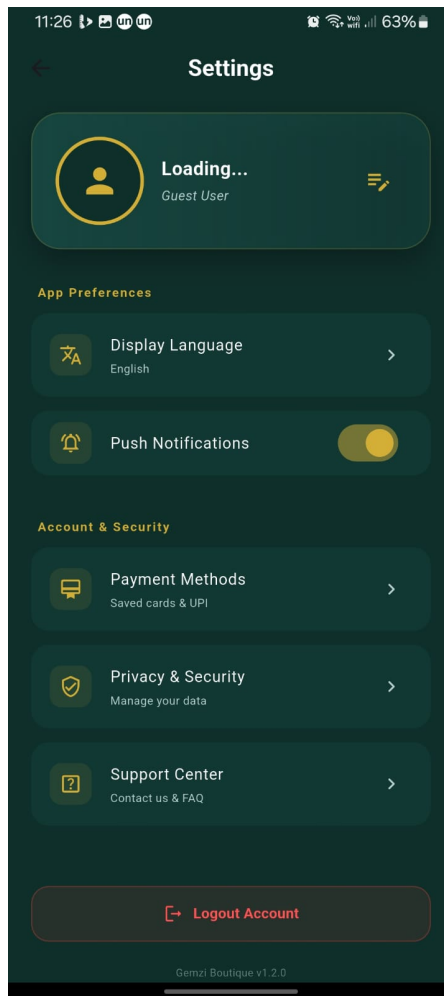


Figure 29: Setting Page

The Settings Page allows users to manage their account and app preferences. It includes options such as profile details, language selection, notifications, payment methods, privacy settings, and support. Users can also log out from this page. It ensures better control, customization, and security of the user account.

10. Admin Dashboard Page

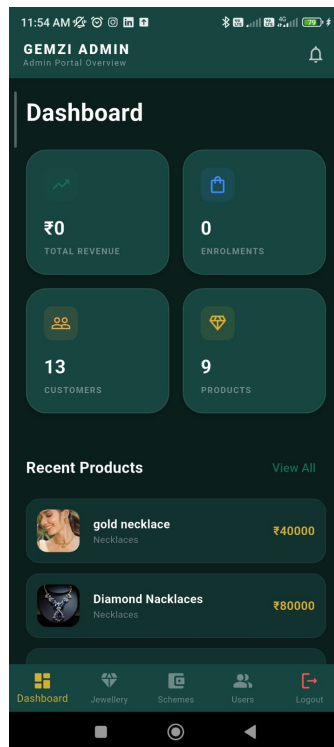


Figure 30: Admin Dashboard Page

The Admin Dashboard Page provides a quick and organized overview of the entire system. It helps administrators efficiently manage products, users, and overall sales activities within the application.

The dashboard displays key information such as total revenue, number of enrollments, customers, and available products, allowing admins to monitor system performance at a glance.

It also includes a recent products section, where newly added items are shown with details like name, category, and price.

4.4 Discussion

Project Workflow and Development Process

The development of the Gemzi Jewellery Marketing and Saving Schemes Application followed a systematic and structured workflow to ensure efficient design, implementation, and user experience. The project began with the requirement analysis phase, where the main objectives of the application were identified. The goal was to create a user-friendly jewellery shopping platform with advanced features such as product browsing, digital gold investment, and augmented reality (AR) try-on. The target users, including general customers and administrators, were also considered to define both user-side and admin-side functionalities.

After gathering requirements, the planning and design phase was carried out. In this stage, the overall architecture of the application was designed, including UI/UX layouts, navigation flow, and feature integration. Wireframes and screen designs were created for key pages such as the login page, home screen, product collections, wallet, AR try-on, and admin dashboard. Special focus was given to designing an attractive and intuitive interface with a consistent color theme to enhance user engagement.

The next stage involved the development phase, where the application was built using Flutter and Dart for cross-platform mobile development. The frontend was developed to provide smooth navigation and interactive user interfaces, while back-end integration (if applicable) was implemented to manage user data, product details, and transactions. Features such as product listing, category filtering, real-time gold rate display, and secure authentication were developed. The AR try-on functionality was implemented to allow users to virtually try jewellery items using the device camera, improving the overall shopping experience.

An important feature implemented in the application is the Add to Cart functionality, which allows users to select and store their preferred jewellery items before making a purchase. Users can easily add products from the collection page, view selected items, and proceed with buying decisions. This feature enhances convenience and improves the overall shopping flow by enabling users to compare and manage multiple items efficiently.

Another key feature is the Language Translation option, which improves accessibility for users from different regions. The application provides language selection support in three languages: English, Hindi, and Marathi, allowing users to view

the app content in their preferred language. This feature makes the application more user-friendly and inclusive, especially for users who are not comfortable with the default language.

Simultaneously, an Admin Panel was developed to manage the system efficiently. The admin dashboard allows administrators to monitor key metrics such as revenue, users, and products. It also provides functionality to add, update, or delete jewellery items, making the system dynamic and easy to maintain.

Following development, the project moved to the testing phase, where different modules were tested to ensure proper functionality and performance. Both manual testing and functional testing were conducted to identify and fix bugs, improve usability, and ensure smooth navigation across all screens. Special attention was given to features like login authentication, AR try-on accuracy, and real-time data display.

Finally, the application entered the deployment and evaluation phase, where the app was prepared for real-world usage. Feedback was analyzed to improve performance, enhance features, and fix any remaining issues. The final product is a complete jewellery application that provides users with a modern, interactive, and convenient shopping experience.

Overall, the project followed a step-by-step development process including analysis, design, development, testing, and deployment, ensuring the successful creation of a reliable and user-friendly application.

CHAPTER 5: Conclusion and Future Scope

5.1 Conclusion

The Gemzi Jewellery Marketing and Saving Scheme Application has been successfully designed and developed as a modern, user-friendly platform that enhances the overall jewellery shopping experience. The application integrates advanced features such as product browsing, digital gold investment, real-time gold rate tracking, and augmented reality (AR) try-on, making it a comprehensive solution for both customers and administrators.

The main objective of this project was to create an application that is not only visually appealing but also highly functional and easy to use. Through proper planning and implementation, the app provides a smooth and efficient interface where users can explore different jewellery categories, view detailed product information, and make informed purchasing decisions. The inclusion of features like Add to Cart, Exclusive Collections, and Search functionality further improves usability and enhances the overall shopping flow.

One of the major highlights of the application is the AR Try-On feature, which allows users to virtually try jewellery items using their device camera. This innovative feature bridges the gap between online and offline shopping by giving users a realistic idea of how the jewellery will look on them. It not only increases user confidence but also improves engagement and satisfaction.

The application also includes a Digital Gold Wallet, where users can invest in gold based on live market rates. This feature provides flexibility and convenience, allowing users to buy gold in small amounts and securely store it digitally. Along with this, the Live Gold Rate Page ensures transparency by displaying up-to-date gold prices, helping users make better financial decisions.

Another important aspect of the application is its multi-language support, which includes English, Hindi, and Marathi. This feature makes the app accessible to a wider audience and ensures inclusivity for users from different linguistic backgrounds.

From an administrative perspective, the Admin Dashboard plays a crucial role in

managing the application. It allows administrators to monitor key metrics such as revenue, customers, and products, and provides the ability to add, update, or remove jewellery items. This ensures that the system remains dynamic, organized, and easy to maintain.

Throughout the development process, various phases such as requirement analysis, design, development, testing, and deployment were carefully followed. Proper testing ensured that the application performs efficiently without major errors, providing a smooth and reliable user experience.

In conclusion, the Gemzi Jewellery Marketing and Saving Schemes Application successfully meets its objectives by combining technology, usability, and innovation. It offers a complete digital solution for jewellery shopping while incorporating advanced features that enhance user interaction and convenience. The project demonstrates how modern mobile applications can transform traditional shopping methods into a more interactive and efficient experience.

5.2 Future Scope

Although the Gemzi Jewellery Marketing and Saving Schemes Application provides a wide range of features, there is still significant scope for future improvements and enhancements to make the system more powerful and user-friendly.

One of the major areas of improvement is the integration of secure online payment systems. In the future, the application can include multiple payment options such as credit/debit cards, UPI, net banking, and digital wallets. This will enable users to complete purchases directly within the app, making the process more seamless and efficient.

The AR Try-On feature can be further enhanced by improving accuracy and adding support for more jewellery types such as full bridal sets, customized fittings, and face detection improvements. Advanced technologies like Artificial Intelligence (AI) can be integrated to provide better alignment and personalized recommendations based on user preferences.

Another potential enhancement is the introduction of a recommendation system, where users can receive personalized suggestions based on their browsing history,

preferences, and previous purchases. This will improve user engagement and help customers discover products more easily.

The application can also be expanded by adding more language options to reach a broader audience across different regions. This will further improve accessibility and make the app more inclusive on a national or even global level.

In terms of the Digital Gold Wallet, future improvements can include features like gold selling, transfer of gold between users, and conversion of digital gold into physical jewellery. This would make the feature more flexible and valuable for users.

Another important future scope is the development of a web-based version or e-commerce website, allowing users to access the platform from desktops and laptops. This will increase the reach of the application and attract more users.

The Admin Panel can also be enhanced with advanced analytics and reporting tools, enabling administrators to track sales trends, user behavior, and product performance more effectively. This will help in better decision-making and business growth.

Security can be further improved by implementing advanced authentication methods such as biometric login (fingerprint/face recognition) and data encryption techniques to ensure user data protection.

Additionally, features like order tracking, reviews and ratings, wishlist, and notifications for offers can be added to enhance user engagement and satisfaction.

In the future, the application can also integrate AI chatbots or customer support systems to provide instant assistance to users, improving the overall user experience.

Overall, the future scope of the Gemzi Jewellery Application is vast, with many opportunities to enhance functionality, improve performance, and expand its reach. With continuous updates and technological advancements, the application can evolve into a complete and advanced jewellery e-commerce platform.

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